

Preparation of electrocompetent cells

Weinstock paper:

Matthew T Weinstock, Eric D Heseck, Christopher M Wilson, Daniel G Gibson

Vibrio natriegens as a fast-growing host for molecular biology

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To prepare the day before:

- LB I media +V2 salts (204 mM NaCl, 4.2 mM KCl, 23.14mM MgCl₂)
- electroporation buffer (680 mM sucrose, 7 mM K₂HPO₄, pH 7) (sterile filtrated)
- over night culture of *V.n* from the cryo-stock in LBI + v2 salts (37 °C ; at 200 r.p.m)

Preparing culture :

Depending on how many aliquds you want to have, incubate media with your overnight culture for a starting OD of 0.05.

The culture is grown at 37 °C in a baffled flask, shaking at 200 r.p.m. until an OD₆₀₀ between 0.5 to 0.8 is reached.

*be careful when they reach an OD near 0.1, *V. natriegens* is very fast growing, so start measuring in shorter time periods.

Prechill the electroporation buffer

Washing:

From here on try always to keep your culture on ice.

- The culture is then put on ice for 15 min.
(the original Protocol sais to directly fill them into your prechilled centrifugatin containments)
- The cells are pelleted at 3000x g. for 20 min at 4 °C.
- The supernatant is carefully decanted and the cell pellets are gently suspended in 10mL of **chilled** electroporation buffer.
- The suspensions are transferred to a **chilled** 50mL falcon tube and the tube is filled top with additional **chilled** electroporation buffer (50mL) and inverted several times.
- The cells are centrifuged down at 3000x g for 15 min at 4 °C.
- The wash is repeated two times for a total of three washes.

Aliquotation:

After the final wash, the supernant is carefully decanted the cells are gently resuspended in residual electroporation buffer.

Measure the OD in a 1/20 dilution against electroporation buffer.

The volume is adjusted with additional electroporation buffer to bring the final OD₆₀₀ to 16.

The Cells are aliquoted (80μL) into chilled 1.5μL centrifugation tubes, directly frozen in liquid nitrogen and stored at -80 °C until use.